

MCQs.

1. C
2. A
3. B
4. D
5. A
6. C
7. D
8. A
9. ~~A~~ B
10. ~~B~~ A
11. B
12. D

need a little bit
adjustment.

and knowledge

1. A ✓

2. B ✓

10. ^{Na}? (Sodium?) or (Avogadro's N.O)?

11. B ✓

12. D ✓

Section - C

Q.

Q. 3 (i) ✓

A. 3 (i) ✓

- Proton's relative charge is +1 and its relative mass is approximately 1 atomic mass unit (amu) or 1.6726×10^{-27} Kg.

Neutron

- ~~Electron~~'s relative charge is 0 and its relative mass is approximately 1 atomic mass unit (amu) or ~~1.6749×10^{-27} Kg~~ or 1.6749×10^{-27} Kg.

- Electron's relative charge is -1 and its relative mass is approximately $\frac{1}{1836}$ or 9.11×10^{-31} Kg.

Location of proton, neutron, and electron in nucleus in atom :-

Protons and neutrons in a nucleus of an atom. whereas, electron orbit around the nucleus of an atom in an energy levels or orbits. They play crucial role in determining properties or behaviour of atoms and molecules. Neutrons and protons are held together by a strong nuclear force. This force exist b/w neutrons-neutrons, protons-protons and neutrons-protons.

Atomic mass number :-

The total number of protons and electron present in nucleus of an atom is called atomic number.

Mass number / nucleon :-

The total number of protons and ~~not~~ neutrons present in an atom is called nucleon or mass number.

Isotopes :-

Isotopes are the atom of an element whose nuclei have same atomic number but different mass number ~~to~~ because of an element can differ in the number of neutrons.

Q.3(ii)

A.3(ii)

Atomic size :-

~~Atom~~ When you move across left to right in a period atomic size decreases because nuclear charge increases while both electrons enter the same shell, increasing the inward pull.

Ionization energy :-

When you move across from left to right in a period ionization energy increases because smaller size and ~~are~~ and greater nuclear charge hold the outer electrons more tightly.

Electronegativity :-

When you move across from left to right in a period, electronegativity increases because atoms attract the ~~shared~~ shared bonding pair ~~more~~ more strongly as charge rises and size falls.

Electropositive

① Electropositive elements lose electrons ~~easily~~ easily and form positive ions (cations).

Electronegative

Electronegative elements gain electrons easily and form negatively charged ions (anions).

Example :-

Na, K

Examples :-

Cl, O.

Q: 3 (iii)

A: 3 (iii)

mole :-

the amount of substances that contains as many particles as there are in atoms exactly 12 g of carbon - 12
i.e. 6.022×10^{23} particles.

Avogadro's number :-

The number of particle in one mole is known as avogadro's number. $= 6.022 \times 10^{23}$ particles.

① moles of H_2SO_4 in 49g.

~~moles :- ?~~
~~molecular mass :- 49g~~
~~molar mass :- H_2SO_4~~
 ~~$2(1) +$~~

moles :- ?
mass :- 49g
m.m. = H_2SO_4
 $2(1) + 32 + 4(16)$
 $= 2 + 32 + 64$
 $= 2 + 96$
 $= 98 \text{ g/mol}$

98mol

moles :- $\frac{\text{mass}}{\text{m.m}}$

$$\frac{49}{98}$$

$$= 0.5 \text{ g/mol}$$

(ii) molecules in 0.5 mol of CO_2

molecules : ?

moles : 0.5

$$N_A = 6.022 \times 10^{23}$$

molecules : moles $\times N_A$

$$0.5 \times 6.022 \times 10^{23}$$

$$= 3.011 \times 10^{23} \text{ molecules of } \text{CO}_2$$

(iii) mass of 2 moles of NaOH .

G. mass :- ?

~~G. mass~~ moles :- 2

m. m : NaOH

$$1(23) + 1(16) + 1(1)$$

$$23 + 16 + 1$$

$$23 + 17$$

40 mole

mass : moles $\times$ m. m

$$= 2 \text{ moles} \times 40 \text{ mol}$$

$$= 80 \text{ g}$$

Q: 3(iv)

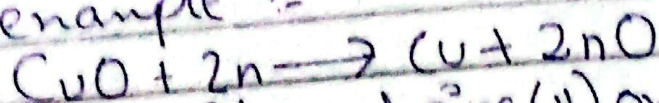
A: 3(iv)

Oxidation - Reduction in transfer of electrons.

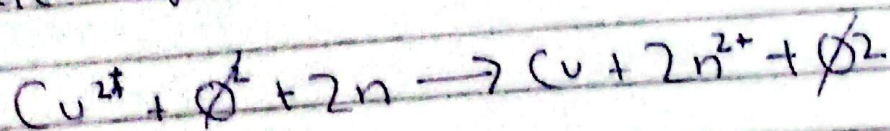
Oxidation:

Oxidation is the loss of electrons
and reduction is the gain of electrons.

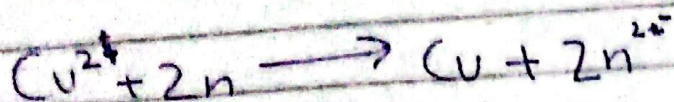
for example :-



Copper (II) oxide and zinc (II) oxide are both ionic compounds. let's rewrite this equation as ionic equation.



Oxide ions are spectator ion, so net equation is :-



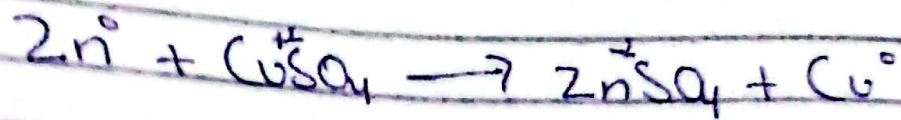
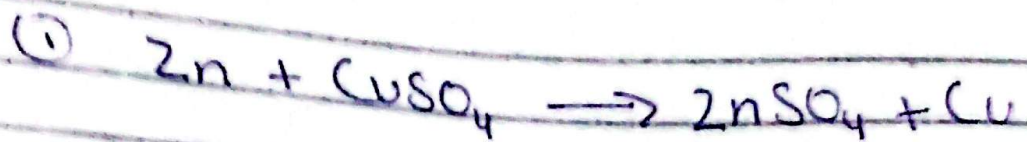
Cu^{2+} ions are gaining 2 electrons to form Cu.
It's oxidation state changes from +2 $\rightarrow$ 0.

So Cu^{2+} are undergoing reduction whereas

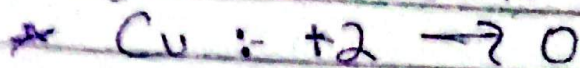
Zn are losing 2 electrons to form Zn^{2+} .

So it's oxidation state changes from 0 $\rightarrow$ +2.

So Zn are undergoing oxidation.



* So it's going to reduced and it's is ~~oxidizing~~ oxidizing agent. because it gains electrons



It's going to oxidized & because it loses it's electrons and that is why it is a reducing agent.

Corrosion :-

is the gradual eating away (oxidation) of a metal surface by reaction with air, moisture or chemicals for iron so it is called rusting or corrosion.

To prevent corrosion :-

The most easiest way to prevent is to paint and galvanizing (zinc coating).

Section - B

Q: 2(i)

A: 2(i)

Chemistry :-

deals with the composition, structure, properties, behaviour, and changes b/w energy and matter.

Branches ~~with~~ of chemistry :-

① Organic Chemistry :-

~~Organo~~ Organic chemistry ~~deals with~~ is the branch of chemistry that deals with the substances containing carbon. (except carbonates, bicarbonates, oxides, carbide).

② Inorganic chemistry :-

Inorganic Chemistry is the branch of chemistry that deals with the elements and compounds except organic compounds.

③ Biochemistry :-

The branch of chemistry that deals with physical and chemical changes occur in living organism is called biochemistry.

Section - B

Q:2(i)

A:2(i)

Chemistry:-

deals with the composition, structure, properties, behaviour, and changes b/w energy and matter.

Branches ~~with~~ of chemistry:-

① Organic Chemistry:-

~~Org~~ Organic chemistry ~~deals with~~ is the branch of chemistry that deals with the substances containing carbon. (except carbonates, bicarbonates, oxides, carbide).

② Inorganic chemistry:-

Inorganic Chemistry is the branch of chemistry that deals with the ~~st~~ elements and compounds except organic compounds.

③ biochemistry:-

The branch of chemistry that deals with physical and chemical changes occur in living organism is called biochemistry.

Q: 2(ii)(or)

A: 2(ii)(or)

Solution	Suspension	Colloid
A homogeneous mixture of 2 or more substance components	A homogeneous mixture of 2 or more components.	A heterogeneous mixture of 2 or more components
② It does not separate	It separates or settle down when stationary	It does not separate.
3) Sea water	muddy water	milk

Q: 2(iii)

A: 2(iii)

Atomic no:-

Atomic number is the total number of protons and electrons present in an ~~atom~~ atom.

Mass number:-

mass number is the total number of neutrons and protons in the nucleus present in the atom.

(i) $^{35}_{17}\text{Cl}$

Protons :- 17

electrons :- 17

neutrons :- 18

Q: 2 (iv) _____
A: 2 (iv) _____

Na :- (11)

$1s^2, 2s^2, 2p^6, 3s^1$
valence electron: 1

Cl :- 17

$1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
valence electron :- 7 or 17

Q: 2 (v) _____
A: 2 (v) _____

Ionization energy :-

It is the minimum energy required to remove the most loosely bound electron from an isolated gaseous atom.

• trend :-

★ Down a group :- It decreases because atomic size increases so the outer electron is further from and less tightly held by the nucleus.

Across of period :-

It increases - nuclear charge rises and size falls, so electrons are held more tightly.

Q: 2(vi) (or) —
formation of $MgCl_2$

* $Mg(2, 8, 2)$ has 2 valence electrons
Chlorine $(2, 8, 7)$ has 7.

* Mg loses its 2 electrons to become Mg^{2+} , each Cl atom gains 1 electron to become Cl^{-}
So 2 Cl atoms are needed.

* The oppositely charged Mg^{2+} and 2 Cl^{-} ions attract electrostatically to form ionic $MgCl_2$.

Q: 2(vii) —
A: 2(vii) —

molecular mass :-

$$1(2) + 32 + 4(16)$$

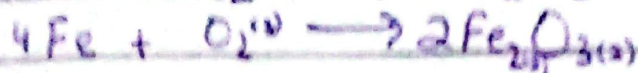
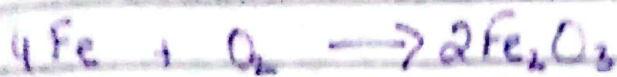
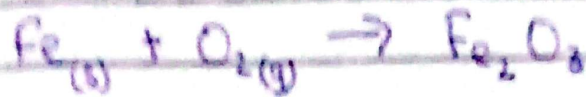
$$2 + 32 + 64$$

$$2 + 96$$

$$98 \text{ g amu.}$$

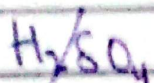
Q: 2(viii)

A: 2(viii)



Q: 2(ix)(or)

A: 2(ix)(or)



~~$1(2) + x + 4(2) = 0$~~

~~$2 + x + 8 = 0$~~

~~$x + 8 + 2 = 0$~~

~~$x + 10 = 0$~~

~~$x = -10$~~

~~$x = -10$~~



$$1(2) + x + 4(-2) = 0$$

$$2 + x - 8 = 0$$

$$x - 8 + 2 = 0$$

$$x - 6 = 0$$

$$x = 0 + 6$$

$$x = +6$$

Q: 2(x)

A: 2(x)

Exothermic reaction :

releases heat to the surrounding. ΔH is negative.
e.g. respiration etc.

Endothermic reactions:

absorbs heat from
surrounding. ΔH is positive e.g.
photosynthesis etc.

Q: 2(ni)(or)

A: 2(ni)(or)

- ① The concentration of reactants or products remain unchanged
- ② The temperature of the system remain ~~to~~ constant.
- ③ The pressure or volume of the system remains constant.